MATH 564 (Monday September 15)

```
case 'BFGS'  %%%%% QUASI-NEWTON (BFGS) %%%%%

    % Compute the BFGS direction.  If iter=1 then use the negative
    % gradient direction.  Also use the negative gradient direction
    % if the update will yield a nearly non-positive definite H.
    if iter==1
        H=max(abs(out.f(1)),sqrt(pr.dftol))*speye(n);   ← initial H is ~I
        p=-H*out.g;                                     ← gradient descent
    else
        s=out.x(:,end)-out.x(:,end-1);
        y=out.g(:,end)-out.g(:,end-1);
        if y'*s>0.0001*norm(s)*norm(y)   ← if sufficiently pos. def.
            r=1/(y'*s);
            I=eye(n);
            H=(I-(r*s)*y')*H*(I-(r*y)*s')+(r*s)*s';
        else                             ← if not then restart H
            H=max(abs(out.f(end)),sqrt(pr.dftol))*speye(n);
        end
        p=-H*out.g(:,end);   ← Newton step
    end

    % call the linesearch in direction p, set alpha=1
    pr.LS.alpha=1;   ← α₀ = 1 is expected with Newton steps
    ox=out.x(:,iter);
    of=out.f(iter);
    og=out.g(:,iter);
    [xnew,flag,nf,ng]=linesearch(ox,of,og,p,pr);
    out.feval=out.feval+nf;
    out.geval=out.geval+ng;
```

PRO

good convergence

CON

memory footprint can be large
computationally intensive
good 2nd order info after n iterations
not applicable for large n

Let's try and "fix" these

we seek a simpler q-Newton update. Consider the simplest possible rank 1 update (ansatz)

$$B_{k+1} = B_k + \underline{\sigma v v^T} \quad \text{symmetric rank 1 update}$$

we don't need $\sigma \in \mathbb{R}$, only the sign matters. So let $\sigma = \pm 1$. This update must satisfy the secant equation.

$$s_k = x_{k+1} - x_k$$
$$y_k = \nabla f_{k+1} - \nabla f_k$$

$$\begin{aligned} y_k &= B_{k+1} s_k \\ &= (B_k + \sigma v v^T) s_k \\ &= B_k s_k + \sigma v v^T s_k \\ &= B_k s_k + (\sigma v^T s_k) v \\ \Rightarrow \quad y_k - B_k s_k &= (\sigma v^T s_k) v \qquad (*) \end{aligned}$$

That is, v is a scalar multiple of $y_k - B_k s_k$. It is a measure of how much the current B_k does not satisfy the secant equation. let

$$v = a(y_k - B_k s_k) \qquad a = (\sigma v^T s_k)^{-1}$$

Substitute into $(*)$:

$$y_k - B_k s_k = \sigma a^2 (y_k - B_k s_k)^T s_k (y_k - B_k s_k)$$

$$\Rightarrow \quad \sigma a^2 (y_k - B_k s_k)^T s_k = 1$$

$$\sigma = \pm 1, \quad a = \pm \left| (y_k - B_k s_k)^T s_k \right|^{-1/2}$$

$$\sigma v v^T = \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}$$

let $w_k = y_k - B_k s_k$

$$B_{k+1} = B_k + \frac{w_k w_k^T}{s_k^T w_k}, \quad w_k = y_k - B_k s_k$$

symmetric rank 1 update
that satisfies the secant equation.

SR1

Why don't we use this instead of BFGS? It is known that this update quickly captures good 2nd derivative info: better than BFGS!

we don't use B, we actually use $H = B^{-1}$ $(p = -H\nabla f)$

Asimilar analysis reveals:

$$H_{k+1} = H_k + \frac{z_k z_k^T}{y_k^T z_k}, \quad z_k = s_k - H_k y_k$$

Ans: No guarantee of positive definiteness! That is, not linesearch criterion can guarantee pos. def. update. We might use $H_{k+1} = H_k + zz^T/y^Tz + \mu I$ for some large enough μ, but then we lose some of the great curvature information we get with SR1.

However, SR1 will turn out to be very useful in a Trust region framework

Trust Region: From iterate x_k, Find $x_{k+1} = x_k + p_k$ the solution to $\min_p m(p)$ s.t. $\|p\| \le \Delta$ for some model $m(p)$ and trust region radius Δ, where $f(x_{k+1})$ shows sufficient decrease.